

GATE TF

Textile Engineering & Fibre Science

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 22 questions (34%)
Moderate: 36 questions (55%)
Difficult: 7 questions (11%)

GENERAL INSTRUCTIONS

GATE TF — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [no -ve]

[General Aptitude]

Perimeter of square side 5:

Enter your answer here

Q2. 1M [no -ve]

[Engineering Mathematics]

Integral of $2x \, dx$:

Enter your answer here

Q3. 1M [-0.33]

[Textile Fibers]

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

Q4. 1M [-0.33]

[Yarn Manufacturing]

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

Q5. 1M [-0.33]

[Fabric Manufacturing]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q6. 1M [-0.33]

[Textile Testing]

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

Q7. 1M [no -ve]

[Chemical Processing]

Reactive dye bonds with:

Enter your answer here

Q8.

1M [-0.33]

[Apparel Engineering]

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

Q9.

1M [-0.33]

[Textile Machinery]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q10.

1M [no -ve]

*[Textile Physics]*Coulomb friction law: $F = ?$

Enter your answer here

Q11.

1M [no -ve]

[General Aptitude]

Perimeter of square side 5:

Enter your answer here

Q12.

1M [-0.33]

[Yarn Manufacturing]

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

Q13.

1M [no -ve]

[Chemical Processing]

Reactive dye bonds with:

Enter your answer here

Q14.

1M [-0.33]

[Textile Machinery]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q15.

1M [-0.33]

[General Aptitude]

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

Q16.

1M [-0.33]

[Textile Machinery]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q17.

1M [-0.33]

[Chemical Processing]

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

Q18.

1M [-0.33]

[Textile Fibers]

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

Q19.

1M [-0.33]

[Apparel Engineering]

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

Q20.

1M [-0.33]

*[Engineering Mathematics]*Integral of $2x \, dx$:

- (A) $x^2 + C$
- (B) $2x + C$
- (C) $x + C$
- (D) $x^2/2$

Q21.

1M [no -ve]

[General Aptitude]

Perimeter of square side 5:

*Enter your answer here***Q22.**

1M [-0.33]

*[Engineering Mathematics]*Integral of $2x \, dx$:

- (A) $x^2 + C$
- (B) $2x + C$
- (C) $x + C$
- (D) $x^{2/2}$

Q23.

1M [-0.33]

*[Textile Physics]*Coulomb friction law: $F = ?$

- (A) μR
- (B) $\mu \omega$
- (C) N/μ
- (D) $\mu' \omega$

Q24.

1M [-0.33]

[Textile Fibers]

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

Q25.

1M [-0.33]

[Fabric Manufacturing]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q26.

1M [-0.33]

[Textile Machinery]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q27.

1M [no -ve]

[Textile Machinery]

Ring frame produces:

*Enter your answer here***Q28.**

1M [-0.33]

[Textile Testing]

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

Q29.

1M [-0.33]

[Fabric Manufacturing]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q30.

1M [-0.33]

[Apparel Engineering]

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

Q31.

2M [no -ve]

[Fabric Manufacturing]

Plain weave has:

*Enter your answer here***Q32.**

2M [-0.66]

[General Aptitude]

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

Q33.

2M [no -ve]

[General Aptitude]

Perimeter of square side 5:

Enter your answer here

Q34.

2M [no -ve]

[Chemical Processing]

Reactive dye bonds with:

*Enter your answer here***Q35.**

2M [no -ve]

[Fabric Manufacturing]

Plain weave has:

*Enter your answer here***Q36.**

2M [-0.66]

[Fabric Manufacturing]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q37.

2M [no -ve]

[Yarn Manufacturing]

Carding does:

*Enter your answer here***Q38.**

2M [-0.66]

[Textile Fibers]

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

Q39.

2M [no -ve]

*[Engineering Mathematics]*Integral of $2x \, dx$:

*Enter your answer here***Q40.**

2M [-0.66]

[General Aptitude]

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

Q41.

2M [-0.66]

[Textile Fibers]

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

Q42.

2M [-0.66]

[Fabric Manufacturing]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q43.

2M [-0.66]

*[Textile Physics]*Coulomb friction law: $F = ?$

- (A) μN
- (B) $\mu \dot{N}$
- (C) N/μ
- (D) $\mu' N$

Q44.

2M [-0.66]

[Textile Fibers]

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

Q45.

2M [no -ve]

[Apparel Engineering]

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

Q46.

2M [-0.66]

[Chemical Processing]

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

Q47.

2M [-0.66]

[Textile Testing]

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

Q48.

2M [no -ve]

[Textile Fibers]

Cotton is a:

*Enter your answer here***Q49.**

2M [no -ve]

[Textile Fibers]

Cotton is a:

*Enter your answer here***Q50.**

2M [-0.66]

[Yarn Manufacturing]

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

Q51.

2M [no -ve]

[Textile Machinery]

Ring frame produces:

*Enter your answer here***Q52.**

2M [-0.66]

*[Textile Physics]*Coulomb friction law: $F = ?$

- (A) μN
- (B) $\mu^2 N$
- (C) N/μ
- (D) $\mu^2 N$

Q53.

2M [-0.66]

[Textile Machinery]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q54.

2M [-0.66]

[Textile Machinery]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q55.

2M [-0.66]

*[Engineering Mathematics]*Integral of $2x \, dx$:

- (A) $x^2 + C$
- (B) $2x + C$
- (C) $x + C$
- (D) $x^2/2$

Q56.

2M [-0.66]

*[Engineering Mathematics]*Integral of $2x \, dx$:

- (A) $x^2 + C$
- (B) $2x + C$
- (C) $x + C$
- (D) $x^2/2$

Q57.

2M [-0.66]

*[Textile Physics]*Coulomb friction law: $F = ?$

- (A) μN
- (B) $\mu \dot{N}$
- (C) N/μ
- (D) $\dot{N}/\mu$

Q58.

2M [-0.66]

[Textile Machinery]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q59.

2M [no -ve]

[Fabric Manufacturing]

Plain weave has:

Enter your answer here

Q60.

2M [-0.66]

*[Textile Physics]*Coulomb friction law: $F = ?$

- (A) ;Ãà
- (B) ;Ãôà
- (C) N/;Ã
- (D) ;Ã' à

Q61.

2M [no -ve]

[Apparel Engineering]

Pattern is for:

*Enter your answer here***Q62.**

2M [no -ve]

[Textile Testing]

Tensile strength measured by:

*Enter your answer here***Q63.**

2M [-0.66]

[Textile Testing]

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

Q64.

2M [-0.66]

[Yarn Manufacturing]

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

Q65.

2M [-0.66]

*[Textile Physics]*Coulomb friction law: $F = ?$

- (A) ;Ãà
- (B) ;Ãôà
- (C) N/;Ã
- (D) ;Ã' à